

MOF_benchwork Project Documentation

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Abstract

MOF_benchwork is a reproducible benchmark pipeline for gas adsorption simulations in metal-organic frameworks. The project currently focuses on Module A, single-component CO₂ adsorption in rigid MOF-5/IRMOF-1, and is designed to grow into mixture adsorption, Widom/Henry calculations, force-field comparisons, and flexible-framework simulations. This document is intended as a living technical and scientific documentation of the project.

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1 Introduction

`MOF_benchmark` aims to provide a reproducible workflow for benchmarking gas adsorption simulations in metal-organic frameworks. The project combines input normalization, material and force-field resolution, LAMMPS input generation, simulation execution, log parsing, and reference-data evaluation.

The first implemented benchmark target is CO₂ adsorption in MOF-5, resolved internally as IRMOF-1. This system is used as the initial Module A case because it is a common reference framework and suitable for validating the pipeline before extending the project to mixtures, Widom insertion, Henry coefficients, or flexible-framework simulations.

1.1 Benchmark Modules

The project is organized around four benchmark modules:

- Module A: single-component adsorption in a rigid framework.
- Module B: mixture adsorption and selectivity.
- Module C: Widom insertion, Henry coefficients, or force-field comparison.
- Module D: flexible-framework simulations or MD/GCMC flexibility studies.

The current documentation focuses on the completed technical state of Module A and lists the remaining scientific validation steps.

2 Related Work

The literature review used for this project identifies four main directions relevant to MOF adsorption benchmarking: classical GCMC simulations for CO₂ capture, active learning for mixture adsorption landscapes, machine-learned interatomic potentials for adsorption Monte Carlo, and flexible-framework workflows coupling molecular dynamics with sorption simulations. These directions motivate the modular structure of `MOF_benchmark` [1, 2, 3, 4].

2.1 GCMC Simulations for Gas Adsorption in MOFs

Grand Canonical Monte Carlo is a standard method for computing adsorption isotherms because it samples particle insertion and deletion at fixed temperature and chemical potential [1, 2, 3]. For adsorption benchmarks, this is especially useful because the simulation directly produces pressure-dependent uptake values that can be compared with reference isotherms.

Tao et al. provide the closest methodological starting point for the current implementation [1]. Their study investigates pure CO₂ adsorption and CO₂/N₂ mixtures in MOF-5, Mg-MOF-74, and ZIF-8. The work is relevant because it combines a classical GCMC setup, explicit pressure ranges, fugacity correction via the Peng-Robinson equation of state, long-range electrostatics, and validation against experimental data. It also reports that GCMC yields absolute adsorption, while experiments often report excess adsorption. This directly motivates the current implementation of both absolute and excess loading in the evaluation pipeline.

2.2 MOF-5 / IRMOF-1 as First Benchmark System

MOF-5, also known as IRMOF-1, is a suitable first benchmark material because it is a well-known rigid framework and appears in literature studies of CO₂ adsorption [1, 5]. Compared with more complex systems such as open-metal-site MOFs or flexible water-responsive MOFs, MOF-5 is a practical baseline for validating the technical pipeline.

The current project therefore starts with CO₂ in MOF-5/IRMOF-1 as Module A. The goal is not yet to solve every adsorption problem, but to establish a reproducible path from configuration to LAMMPS input generation, execution, parsing, and reference comparison.

2.3 Force Fields and Charge Models

Classical force fields are computationally efficient and make long Monte Carlo simulations feasible. However, the reviewed literature also shows that their reliability depends strongly on the material and adsorbate. UFF or DREIDING-style force fields may be sufficient for initial rigid-framework benchmarks, but they can become unreliable for open metal sites, strongly polar adsorbates, cooperative interactions, or flexible frameworks [1, 3, 4].

This motivates the current design decision to keep force-field resolution explicit and reproducible. For Module A, the framework is treated as rigid, framework charges are resolved from the selected CIF charge scheme, and the CO₂ adsorbate model is read from CRAFTED/RASPA-style definition files. This keeps the first benchmark simple enough to debug while preserving the information needed for later force-field comparisons [5, 3].

2.4 Reference Data and Adsorption Basis

Reference data are not automatically comparable just because they describe the same adsorbent and adsorbate. Relevant differences include temperature, pressure grid, material naming, charge scheme, force field, and adsorption basis. The adsorption basis is particularly important: simulation output from GCMC molecule counts is naturally absolute adsorption, while experimental sources may report excess adsorption [1, 7].

The current project therefore records the reference basis where possible and evaluates references by basis: absolute or simulation-reference data are compared against absolute loading, excess references are compared against excess loading, and unknown-basis references are explicitly marked. CRAFTED is used as a structured simulation-reference source, while NIST ISODB is used as an additional literature and experimental reference source [5, 6, 7].

2.5 Mixture Adsorption and Active Learning

Mixture adsorption introduces a larger thermodynamic feature space because pressure, composition, and temperature can vary simultaneously. Mukherjee et al. show that active learning can reduce the number of required GCMC simulations for multicomponent adsorption landscapes [2]. Their work uses GCMC data as reference data and applies Gaussian process regression to select new simulation points based on model uncertainty.

This motivates Module B as a future extension. After the single-component baseline is validated, the benchmark can be extended to CO₂/N₂ mixtures and later to reduced sampling strategies. In that stage, relevant metrics will include component uptake, selectivity, mean relative error, coefficient of determination, and number of GCMC points saved [1, 2].

2.6 Machine-Learned Interatomic Potentials

Edwards et al. address the question of potential quality in adsorption Monte Carlo [3]. Their MLIP-MC framework enables Widom insertion and GCMC calculations using machine-learned interatomic potentials. The central conclusion from the literature review is that universal MLIPs are not automatically reliable for quantitative adsorption predictions. Some models overbind and fill pores too strongly, while others underpredict adsorption.

This motivates Module C as a potential benchmark rather than an assumption that MLIPs are always better. The project should eventually compare classical force fields, universal MLIPs, adsorption-relevant MLIPs, and reference models using quantities such as Widom insertion energy, Henry coefficients, isosteric heat of adsorption, isotherm error, and runtime [3].

2.7 Framework Flexibility

Rigid-framework GCMC is not sufficient for every MOF. Wang et al. show that water adsorption in NbOFFIVE-1-Ni requires a flexible-framework treatment because guest molecules affect the framework structure [4]. Their workflow couples MLIP-based molecular dynamics with GCMC sorption steps, allowing the framework to relax in response to adsorbate loading.

This motivates Module D as an advanced extension. It should come after the rigid-framework pipeline is validated because it requires additional infrastructure for structural states, molecular dynamics, MLIP models, trajectory analysis, and convergence across sorption-relaxation cycles [4].

2.8 Research Gap and Project Motivation

The reviewed literature shows that many simulation methods and datasets already exist, but reproducible benchmarking remains difficult. A practical benchmark must specify the material, structure preparation, charge model, force field, adsorbate model, fugacity or chemical potential conversion, simulation length, pressure range, reference data, adsorption basis, and uncertainty treatment [1, 2, 3, 4, 5, 7].

MOF_benchmark addresses this gap by turning a benchmark JSON file into a resolved run plan, generated LAMMPS inputs, parsed results, and curated reference comparisons. The current Module A implementation is therefore the first reproducible baseline from which mixture, potential, and flexibility benchmarks can be developed.

3 Project Architecture

The project separates planning, materialization, execution, parsing, and evaluation. This separation is important because the benchmark should first produce a reproducible run plan before any files are written or LAMMPS is started.

Path	Purpose
benchmark.py	Command-line entry point and high-level orchestration.
pipeline/config.py	Input loading, schema normalization, and default values.
pipeline/planning.py	Resolution of material, force field, references, unit cells, and run-plan construction.
pipeline/prepare.py	Side-effect-free preparation plan for generated files.
pipeline/materialization.py	Writes the prepared benchmark environment to disk.
pipeline/runners.py	Executes LAMMPS runs and triggers evaluation.
pipeline/evaluate.py	Converts simulation summaries to CSV, Markdown reports, and plots.
pipeline/eos.py	Equation-of-state helper functions for fugacity and gas density.
converter/cif_to_lammps_data.py	Converts CIF framework structures to LAMMPS data files.
converter/molecule_to_lammps_templates.py	Converts CRAFTED/RASPA molecule definitions to LAMMPS molecule templates.
converter/forcefield_to_lammps.py	Converts force-field parameters to LAMMPS pair coefficients.
resolvers/	Resolves materials, force fields, unit cells, references, and task modules.
parsers/lammps_log_parser.py	Parses LAMMPS thermo output and summarizes adsorption.
input_json_files/	Benchmark input configurations.

3.1 Pipeline Stages

1. **Configuration:** load and normalize the benchmark JSON file.
2. **Resolution:** resolve material aliases, CIF paths, force-field files, unit cells, and references.
3. **Planning:** build a JSON-serializable run plan without side effects.
4. **Preparation:** create a description of required generated files.
5. **Materialization:** write data, templates, force-field includes, inputs, and source copies to disk.
6. **Execution:** run a smoke test or full pressure-grid isotherm in LAMMPS.
7. **Parsing:** extract adsorption summaries from LAMMPS logs.
8. **Evaluation:** calculate absolute and excess loading, compare to references, and generate reports.

4 Benchmark Configuration

Benchmark inputs are stored as JSON files in:

```
input_json_files/
```

The main Module A paper-like configuration is:

```
input_json_files/benchmark_tao2022_mof5_co2.json
```

4.1 Configuration Sections

- **material:** framework name, CIF source, charge scheme, and source dataset.
- **adsorbates:** adsorbate components and optional mixture definition.
- **conditions:** temperature and pressure grid.
- **benchmark:** task, metrics, paper metadata, and reference-data configuration.
- **simulation:** LAMMPS/GCMC settings.
- **evaluation:** absolute/excess handling and gas-density model.
- **output:** output directory, run ID, overwrite behavior, and saved artifacts.

4.2 Evaluation Configuration

The current evaluation block is:

```
"evaluation": {  
  "simulation_basis": "absolute",  
  "report_excess": true,  
  "pore_volume_cm3_g": "auto",  
  "pore_volume_source": null,  
  "pore_volume_method": null,  
  "gas_density_backend": "HEOS",  
  "reference_matching": "by_basis"  
}
```

With `pore_volume_cm3_g` set to `auto`, the run planner reads the accessible pore volume from CRAFTED for the resolved material. For IRMOF-1, this currently resolves to:

$$V_{\text{pore}} = 0.845545 \text{ cm}^3 \text{ g}^{-1}. \quad (1)$$

4.3 Output Directories

Runs are written below:

`outputs/module_A_co2_isotherm/`

To avoid overwriting older results, production runs should use either `-new-run` or a unique `-run-id`.

5 Module A: Single-Component Adsorption

Module A targets single-component adsorption in a rigid framework. The current implementation uses CO2 adsorption in MOF-5/IRMOF-1 as the first benchmark case.

5.1 Current Scope

Module A currently covers:

1. loading a benchmark JSON input file,
2. resolving MOF-5 to IRMOF-1,
3. resolving the DDEC CIF file,
4. resolving CRAFTED UFF force-field files,
5. resolving CO2 molecule definitions,
6. generating a LAMMPS framework data file,
7. generating a CO2 LAMMPS molecule template,
8. generating pair coefficients and LAMMPS force-field include files,
9. generating one GCMC input per pressure point,
10. executing LAMMPS pressure-point simulations,
11. parsing LAMMPS logs,
12. calculating absolute and excess adsorption,
13. comparing the simulation results against reference data.

5.2 Current Paper-Like Setup

Setting	Current value
Material	MOF-5, resolved as IRMOF-1
Charge scheme	DDEC
Adsorbate	CO2
Temperature	298.15 K

Pressure points	0.1, 0.5, 1, 2, 5, 10, 20, 35 bar
Force field	UFF framework and CRAFTED CO2 definition
Framework model	Rigid
Unit cells	1 x 1 x 1
LAMMPS units	real
Pair style	lj/cut/coul/long
Cutoff	12.8 Angstrom
KSpace	pppm, accuracy 1e-4
Fugacity model	Peng-Robinson via CoolProp
Production cycles	500000
Initialization cycles	500000

5.3 CIF to LAMMPS Data

The CIF converter reads the framework geometry with ASE and separately parses CIF charges from `_atom_site_charge`. This is necessary because ASE reads the geometry and cell but does not reliably preserve the charge column required for LAMMPS `atom_style full`.

The generated framework data file contains:

- atom count,
- atom types,
- restricted triclinic box parameters,
- masses,
- `Atoms # full` section with molecule ID, atom type, charge, and Cartesian coordinates.

For IRMOF-1, the framework contains 106 atoms in the primitive simulation cell.

5.4 Molecule Templates

The molecule converter parses CRAFTED/RASPA `.def` files for adsorbates such as CO2 and N2. For CO2, it creates a LAMMPS molecule template with:

- three atoms,
- two rigid bonds,
- global atom type IDs,
- partial charges from the pseudo-atom definitions.

Because the CO2 template contains bonds, LAMMPS input generation reserves additional per-atom bond and special-neighbor slots:

```
extra/bond/per/atom 2 extra/special/per/atom 2
```

5.5 Force-Field Conversion

The force-field converter parses CRAFTED UFF files and writes LAMMPS pair coefficients. It assigns atom types for framework atoms and adsorbate atoms, then applies Lorentz-Berthelot mixing for cross interactions.

The generated force-field include file is referenced by every generated LAMMPS input script.

5.6 LAMMPS Execution

A short smoke test can be started with:

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-test
```

A full pressure-grid run can be started with:

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-isotherm --new-run --jobs 4
```

The `-new-run` flag creates a timestamped output directory. The `-jobs` flag controls how many pressure points are executed in parallel.

5.7 GCMC Interpretation

In a GCMC simulation, CO₂ molecules are inserted and deleted during the simulation. Therefore, the number of CO₂ molecules in a trajectory or dump file changes over time. This is expected behavior and is the basis for estimating adsorption from the mean molecule count.

5.8 Visualization

LAMMPS trajectory dumps are written for pressure-point runs when dump output is enabled in the generated GCMC input. The dump files can be opened with the OVITO desktop application. Typical dump location:

```
outputs/module_A_co2_isotherm/runs/<run_id>/dumps/gcmc_10bar.lammpstrj
```

Coloring by particle type is useful to distinguish framework atom types from inserted CO₂ atom types.

6 Reference Data Curation

The current workflow supports two reference sources:

- CRAFTED as the primary simulation-reference dataset.
- NIST ISODB as an additional experimental or literature reference source.

Reference curation is important because not every available isotherm is directly comparable to a simulation result. Potential problems include different adsorption basis, different temperature, different pressure range, different material naming, duplicated literature entries, and outlier datasets.

6.1 CRAFTED References

CRAFTED references are used as structured benchmark references where available. For the current MOF-5/IRMOF-1 and CO₂ workflow, they provide the primary comparison dataset and geometric descriptors such as accessible pore volume.

6.2 NIST ISODB References

NIST ISODB entries are searched by material aliases, adsorbate, temperature, and pressure overlap. The resolver ranks candidate references and records metadata such as DOI, source, adsorption basis, and exclusion reasons.

Candidate references can be excluded from active evaluation if they are detected as outliers or if their adsorption basis is not allowed by the benchmark configuration. Excluded entries are documented in:

`evaluations/reference_candidates.json`

6.3 Adsorption Basis

The reference parser distinguishes:

- `simulation_reference`,
- `absolute`,
- `excess`,
- `unknown`.

The current evaluation compares references by adsorption basis:

- simulation-reference and absolute references are compared to absolute simulation loading,
- excess references are compared to excess simulation loading,
- unknown-basis references are reported as unknown and compared to absolute loading by default.

6.4 Outlier Handling

Outliers should not be silently removed. Instead, the resolver stores excluded references and the reasons for exclusion. This makes the reference curation reproducible and easier to discuss scientifically.

7 Evaluation and Statistics

The evaluation layer converts LAMMPS log summaries into adsorption isotherm tables, reports, and plots. The simulation molecule count is interpreted as absolute adsorption, and excess adsorption can be calculated when pore volume and gas density are available.

7.1 Absolute Adsorption

The absolute loading is calculated from the mean number of adsorbate molecules in the simulation cell:

$$n_{\text{abs}} = \frac{1000 \cdot \langle N_{\text{ads}} \rangle}{m_{\text{framework}}} \quad (2)$$

where:

- n_{abs} is the absolute adsorption loading in mol kg^{-1} ,

- $\langle N_{\text{ads}} \rangle$ is the mean number of adsorbate molecules per simulation cell,
- $m_{\text{framework}}$ is the framework mass in atomic mass units.

The numerical values of mol kg^{-1} and mmol g^{-1} are identical.

7.2 Excess Adsorption

Excess adsorption is calculated as:

$$n_{\text{ex}} = n_{\text{abs}} - \rho_{\text{gas}} V_{\text{pore}} \quad (3)$$

where:

- n_{ex} is the excess adsorption loading in mol kg^{-1} ,
- n_{abs} is the absolute adsorption loading in mol kg^{-1} ,
- ρ_{gas} is the bulk gas molar density in mol m^{-3} ,
- V_{pore} is the pore volume in $\text{m}^3 \text{ kg}^{-1}$.

For IRMOF-1, the current automatic pore volume source is:

CRAFTED-2.0.0/RAC_DBSCAN/CRAFTED_MOF_geometric.csv:AV_cm^3/g

with:

$$V_{\text{pore}} = 0.845545 \text{ cm}^3 \text{ g}^{-1} = 0.000845545 \text{ m}^3 \text{ kg}^{-1}. \quad (4)$$

The gas density is calculated with CoolProp using the configured backend. For production evaluation, the current backend is HEOS.

7.3 Current Evaluation Outputs

For each evaluated run, the pipeline writes:

- `evaluated_isotherm.csv`,
- `evaluation_report.md`,
- `isotherm_comparison.png`.

For comparison against all available references, the pipeline writes:

- `evaluations/reference_candidates.json`,
- `evaluations/combined_reference_comparison.csv`,
- `evaluations/combined_reference_comparison.png`,
- one evaluation subdirectory per active reference.

7.4 Planned Statistical Extensions

The next scientific development step is statistical validation. The most important additions are:

- convergence analysis by plotting loading as a function of simulation step or block index,
- multiple independent runs using different random seeds,
- standard deviation across seeds,
- block standard error of the mean,
- MAE and MSE per reference dataset,
- automatic highlighting of the best-matching reference.

8 Usage

The recommended environment is the Conda environment `MOF_sim`. It should contain Python, ASE, LAMMPS, CoolProp, matplotlib, and optional OVITO support.

8.1 Dry Run

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --dry-run
```

This prints the resolved run plan and does not write files.

8.2 Prepare Files

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --prepare --new-run
```

This materializes the benchmark directory with generated LAMMPS inputs and copied source files.

8.3 Run Smoke Test

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-test --new-run
```

This runs a short GCMC test to verify that the generated LAMMPS files are readable and that insertion can occur.

8.4 Run Full Isotherm

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --run-isotherm --new-run --jobs 4
```

The `-jobs` option parallelizes pressure points. It should be chosen based on available CPU resources and memory.

8.5 Evaluate Existing Run

```
conda activate MOF_sim
python benchmark.py benchmark_tao2022_mof5_co2.json --evaluate-all --run-id <run_id>
```

This evaluates an already completed run against all active reference datasets.

9 Validation Status

The current test suite checks:

- benchmark JSON loading,
- default configuration normalization,
- MOF-5 to IRMOF-1 alias resolution,
- automatic module selection,
- CRAFTED molecule parsing,
- CIF charge parsing,
- LAMMPS data generation,
- force-field parsing and pair-coefficient generation,
- GCMC input generation,
- LAMMPS log parsing,
- isotherm CSV generation,
- NIST reference parsing,
- combined reference evaluation,
- excess adsorption comparison against excess references.

The latest local test run completed with:

```
Ran 44 tests in 41.431s
OK (skipped=7)
```

The skipped tests depend on optional local tools such as ASE, LAMMPS, or the local NIST ISODB checkout.

9.1 Environment Notes

CoolProp was verified in the `MOF_sim` environment for CO₂ at 298.15 K and 1 bar. The calculated molar gas density was:

$$\rho_{\text{CO}_2} = 40.54118025466975 \text{ mol m}^{-3}. \quad (5)$$

The base Python interpreter may not contain all simulation dependencies. Production commands should therefore be run from `MOF_sim`.

10 Limitations and Future Work

Module A is technically runnable, but the scientific validation is not complete yet. The most important remaining limitations are:

- convergence has not yet been quantified,
- uncertainty estimates are not yet calculated,
- multiple independent random seeds are not yet automated,
- block averaging is not yet implemented,
- MAE and MSE per reference should be made more explicit in a combined report,
- the best-matching reference should be highlighted automatically.

10.1 Immediate Next Steps

1. Add convergence analysis by plotting loading as a function of simulation step or block index.
2. Add multiple-seed execution for every pressure point.
3. Calculate statistical uncertainty using standard deviation across seeds or block standard error of the mean.
4. Refactor the combined evaluation report to show MAE and MSE per reference.
5. Highlight the best-matching reference dataset.
6. Add a final Module A report that combines input protocol, simulation results, reference comparison, uncertainty, and plots.

10.2 Scientific Completion Criteria for Module A

Module A can be considered scientifically complete when:

- the simulation protocol is fully documented,
- adsorption isotherms are converged or convergence limitations are explicitly stated,
- uncertainty is reported for every pressure point,
- multiple seeds or statistically justified block averages are available,
- references are curated and basis-matched,
- absolute and excess adsorption are reported where appropriate,
- generated outputs are reproducible from a single benchmark configuration.

11 Future Modules

The architecture is intended to support additional benchmark modules without duplicating the central planning and materialization logic. Shared functionality such as configuration loading, reference resolution, force-field conversion, execution, and evaluation should remain in `pipeline/`, `converter/`, `resolvers/`, and `parsers/`. Module-specific code should only describe module-specific scientific workflows.

11.1 Module B: Mixture Adsorption and Selectivity

Module B will extend the current single-component workflow to mixtures such as CO₂/N₂. Expected additions include:

- mixture composition in the benchmark configuration,
- multiple molecule templates,
- mixture GCMC input generation,
- component-wise adsorption summaries,
- selectivity calculation,
- reference comparison for mixture data where available.

11.2 Module C: Widom Insertion and Henry Coefficients

Module C will focus on low-pressure adsorption properties and potential comparison. Expected additions include:

- Widom insertion input generation,
- Henry coefficient extraction,
- optional force-field comparison,
- low-pressure reference matching,
- uncertainty estimation for insertion averages.

11.3 Module D: Framework Flexibility

Module D will address flexible-framework simulations. Expected additions include:

- flexible framework preparation,
- MD or hybrid MD/GCMC workflows,
- trajectory analysis,
- OVITO-based visualization outputs,
- comparison between rigid and flexible adsorption behavior.

11.4 Avoiding Redundancy

Preparation and materialization should remain central whenever possible. Individual modules should not duplicate generic logic such as copying files, creating run directories, resolving references, or writing evaluation outputs. Instead, modules should provide the module-specific parts of the run plan that the central pipeline can materialize and execute.

References

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